

# CIS 371 Web Application Programming

## Styles in CSS

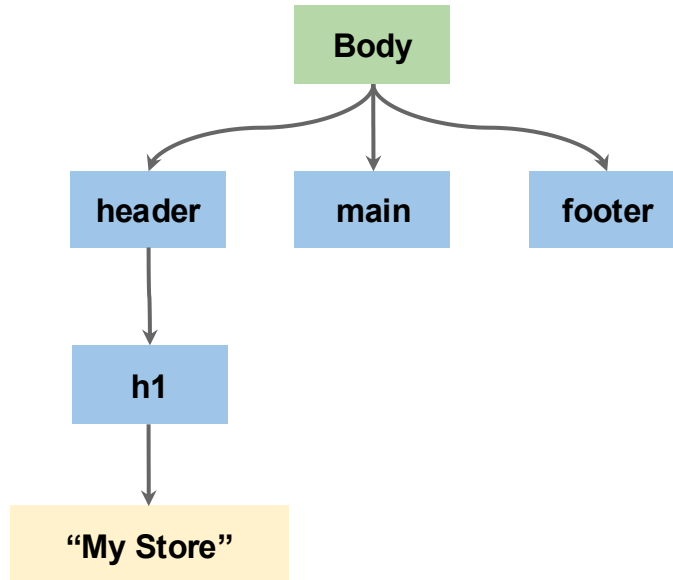


Lecturer: **Dr. Haoyu Li**



# Exercise from the previous class

## Practice: HTML Coding with Emmet



Answer

# The Origins of Cascading Style Sheets

browser wars 1990s



World Wide Web Consortium



universal access



## CROSS-BROWSER TESTING



|                                                                                   |                 |     |
|-----------------------------------------------------------------------------------|-----------------|-----|
|  | Google Chrome   | 50% |
|  | Mozilla Firefox | 30% |
|  | Microsoft Edge  | 20% |

cross browser compatibility problem



HTML: structure/content CSS: presentation/style

# History of CSS

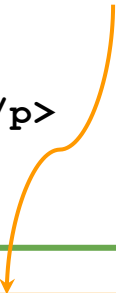
## 20 Years of CSS

<https://www.w3.org/Style/CSS20/>

CSS Zen Garden launched in 2003. <http://www.csszengarden.com/>

# Applying CSS to HTML

```
<html>                                Option 1: External
<head>
  <link rel="stylesheet"
        href="mystyles.css">
</head>
<body>
  <p>Hello world</p>
</body>
</html>
```



```
/* in mystyles.css */
p {
  border: 2px
  solid red;
}
```

```
<html>                                Option 2: Internal
<head>
  <style>
    p {
      border: 2px solid red;
    }
  </style>
</head>
<body>
  <p>Paragraph 1</p>
  <p>Paragraph 2</p>
</body>
</html>
```

```
<body>                                Option 3: Inline (Not recommended)
  <!-- inline style -->
  <p style="border: 2px solid red">.....</p>
</body>
```

# Cascading Order



If we apply CSS to an HTML element using external, internal, and inline methods simultaneously, and they conflict with each other, which CSS style will take precedence?

# Cascading Order

- Situation: A browser is presented with a number of CSS statements, some of which conflict with each other.
- All the styles will "cascade" into a new "virtual" style sheet by the following rules:

- |                         |                                                                                     |         |
|-------------------------|-------------------------------------------------------------------------------------|---------|
| 1. Browser Default      |  | Lowest  |
| 2. External Style Sheet |                                                                                     |         |
| 3. Internal Style Sheet |                                                                                     |         |
| 4. Inline Style         |                                                                                     | Highest |

- Rule 4 has the highest priority

**Specific CSS rules overrule general ones.**

# How to define styles?

- Styles are defined using a set of rules
- Each rule
  - begins with a selector to select the element(s) onto which the rule is applied.
  - Specifies a group of properties to apply to the element(s).

```
selectorA {                                Rule 1
    property1: value;
    property2: value;
}

selectorB {                                Rule 2
    property1: value;
    property2: value;
}
```

**Complete list of CSS properties:**

<https://www.w3.org/Style/CSS/all-properties.en.html>



# Too many CSS properties to memorize



# Too many CSS properties to memorize

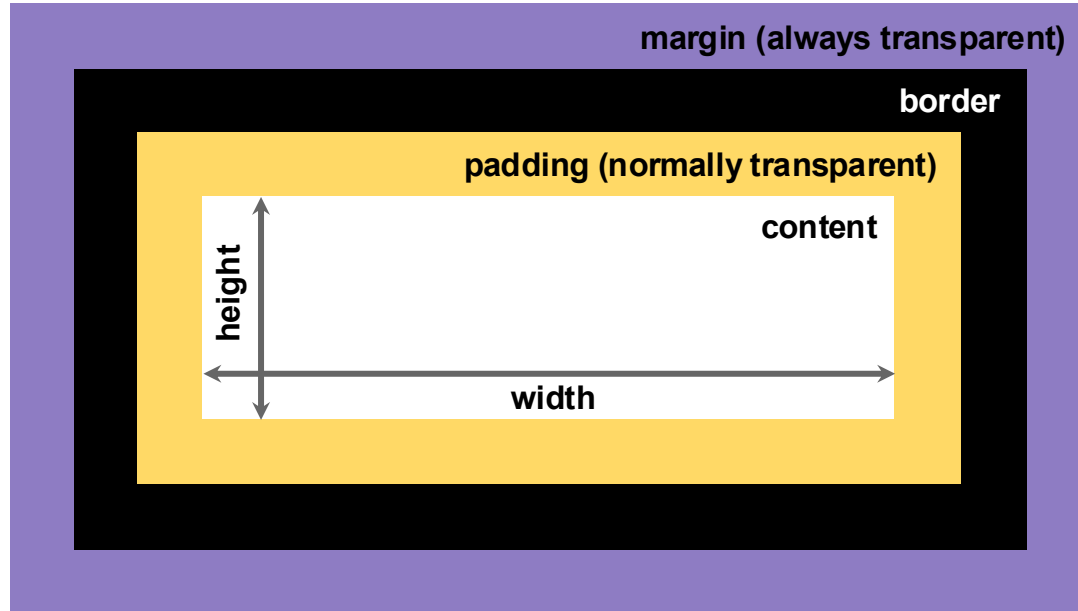


Tip

Use VSCode suggested completion to help you find what you are looking for!



# CSS Box Model



**background-color paints the content, padding, and border**

# CSS Box Model: Padding (inside the border)

```
span {  
  padding: 4px;  
  border: 12px solid green;  
  background: beige;  
}
```

```
<span>Sample Text</span>
```

Sample Text

```
span {  
  padding: 16px;  
  border: 12px solid green;  
  background: beige;  
}
```

Sample Text

# CSS Box Model: Margin (outside the border)

```
span {  
  margin-right: 2px;  
  border: 8px solid green;  
  background: beige;  
}
```

```
<span>Sample</span> Text
```

Sample Text

```
span {  
  margin-right: 8px;  
  border: 4px solid green;  
  background: beige;  
}
```

Sample Text

# CSS Colors

|             |        |             |
|-------------|--------|-------------|
| DeepSkyBlue | Gold   | DarkRed     |
| Beige       | Orange | ForestGreen |
| HotPink     | Olive  | YellowGreen |

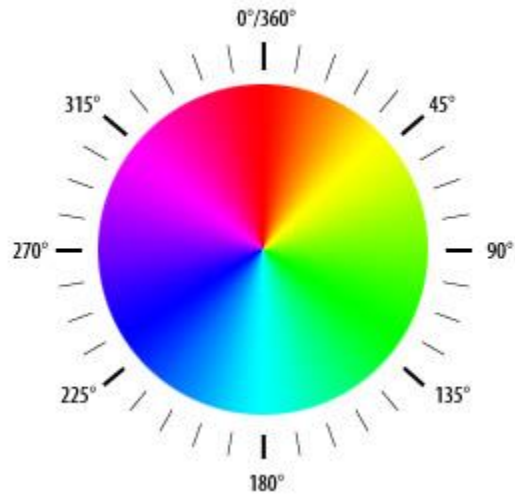
**140 standard names**

[https://www.w3schools.com/colors/colors\\_names.asp](https://www.w3schools.com/colors/colors_names.asp)

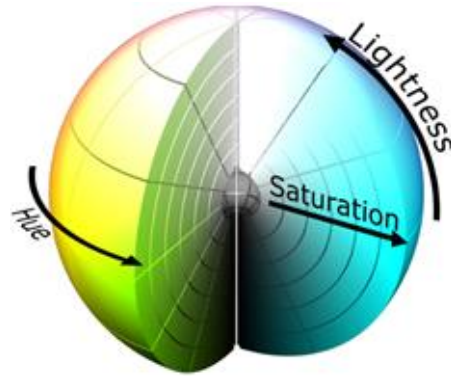
# CSS Colors

- **RGB** (0-255 per color) & **Alpha Transparency**
  - `rgb(155, 138, 73)`
  - `rgba(155, 138, 73, 0.6)`
- **Hex String** (00-FF per color)
  - `#C55` or `#FCA9`
  - `#9B8A49` or `#9B8A493F`
  - **A-10, B-11, C-12, D-13, E-14, F-15**
- **HSL**
  - `hsl(20, 85%, 30%)` or `hsla(20, 85%, 35%, 0.7)`
  - Benefit: easy to generate shades of a particular color (**in code**)

# HSL Colorspace



Hue



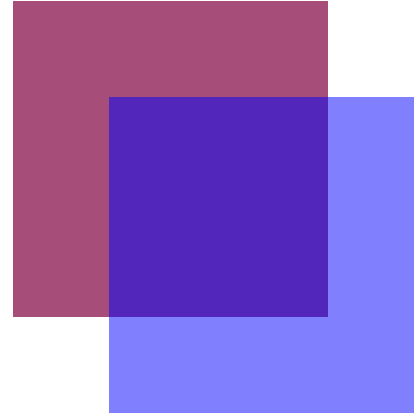
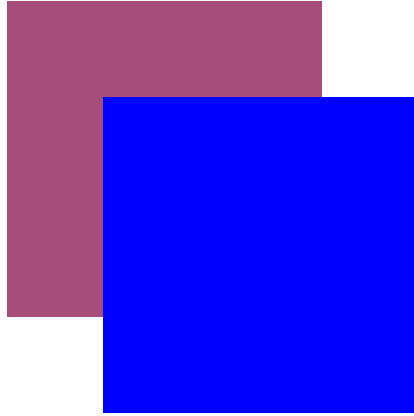
|                   | Description                                             | Range of values                             |
|-------------------|---------------------------------------------------------|---------------------------------------------|
| <b>Hue</b>        | Color Tone                                              | Red:0, Green:120, Blue:240                  |
| <b>Saturation</b> | How much “ink” in your paint                            | 0%: no ink, 100%: max ink                   |
| <b>Lightness</b>  | How much light available when you are viewing the color | 0%: no light 100%: infinite amount of light |

[HSL Color Picker \(CodePen\)](#)

[YUI HSL Color Picker](#)

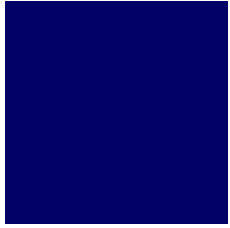


# Color Transparency

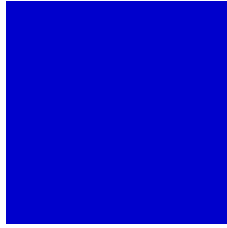


|                   | Opaque Blue                           | 50% Transparent Blue                  |
|-------------------|---------------------------------------|---------------------------------------|
| <b>RGB</b>        | <code>rgb(0, 0, 255, 1.0)</code>      | <code>rgb(0, 0, 255, 0.5)</code>      |
| <b>Hex String</b> | <code>#0000FFFF</code>                | <code>#0000FF7F</code>                |
| <b>HSL</b>        | <code>hsl(240, 100%, 50%, 1.0)</code> | <code>hsl(240, 100%, 50%, 0.5)</code> |

# HSL Practical Use: shade of color tones



**hsl(240, 100%, 20%)**



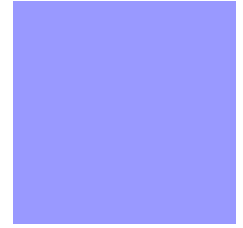
**hsl(240, 100%, 40%)**



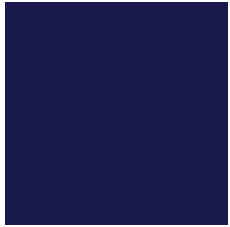
**hsl(240, 100%, 60%)**



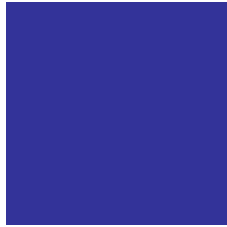
**hsl(240, 100%, 70%)**



**hsl(240, 100%, 80%)**



**hsl(240, 50%, 20%)**



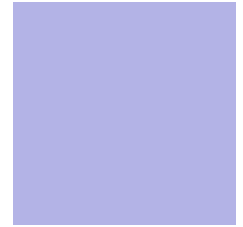
**hsl(240, 50%, 40%)**



**hsl(240, 50%, 60%)**



**hsl(240, 50%, 70%)**

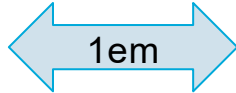


**hsl(240, 50%, 80%)**

# Font Size: 1em

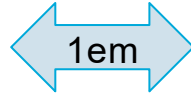
Font: Syncopate

M



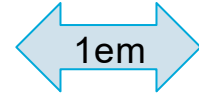
Font: Roboto

M



Font: Lobster

*M*



**1em: the width of uppercase M in the current font (traditional interpretation)**

**1em: the width of the current font (modern typography interpretation)**

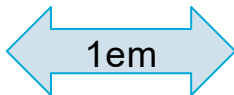
1 em: relative to the nearest parent's font

1 rem: relative to the root font

# Font Size: 1em

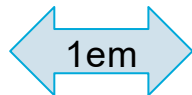
Font: Syncopate

M



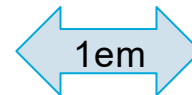
Font: Roboto

M



Font: Lobster

*M*



**1em: the width of uppercase M in the current font (traditional interpretation)**

**1em: the width of the current font (modern typography interpretation)**

1 em: relative to the nearest parent's font

1 rem: relative to the root font

Use “em” for setting spacing around  
your text

# Applying CSS to HTML

| HTML                                                                                             | CSS                                                  | Scope of Application                                              |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------|
| <pre>&lt;!-- by unique id --&gt; &lt;tag id="ticket"&gt;content&lt;/tag&gt;</pre>                | <pre>#ticket {     padding-left: 2em; }</pre>        | Only to one element <code>#ticket</code>                          |
| <pre>&lt;!-- by tag name --&gt; &lt;xyz&gt;content&lt;/xyz&gt;</pre>                             | <pre>xyz {     font-weight: bold; }</pre>            | All <code>&lt;xyz&gt;</code> tags in the document                 |
| <pre>&lt;!-- by class name --&gt; &lt;xyz class="weekend"&gt;content&lt;/xyz&gt;</pre>           | <pre>.weekend {     border: 2px solid brown; }</pre> | All <code>.weekend</code> class in the document                   |
| <pre>&lt;!-- by other attributes --&gt; &lt;xyz anyattr="somevalue"&gt;content&lt;/xyz&gt;</pre> | <pre>[anyattr] {     background-color: white }</pre> | All tags with this attribute <code>anyattr</code> in the document |



# Example: Using CSS Selectors

```
<html>
  <head>
    <link rel="stylesheet" href="mystyles.css">
  </head>
  <body>
    <span lang="en">Hello World</span>
    <p>I am learning
      <span id="abbrev">CSS</span>
    </p>
  </body>
</html>
```

```
#abbrev {
  border-color: red
}
```

Hello World

I am learning CSS

# Example: Using CSS Selectors

```
<html>
  <head>
    <link rel="stylesheet" href="mystyles.css">
  </head>
  <body>
    <span lang="en">Hello World</span>
    <p>I am learning
      <span id="abbrev">CSS</span>
    </p>
  </body>
</html>
```

```
#abbrev {
  border-color: red
}
```

Hello World

I am learning CSS

# Example: Using CSS Selectors

```
<html>
  <head>
    <link rel="stylesheet" href="mystyles.css">
  </head>
  <body>
    <span lang="en">Hello World</span>
    <p>I am learning
      <span id="abbrev">CSS</span>
    </p>
  </body>
</html>
```

```
span {
  border-color: red
}
```

Hello World

I am learning CSS



# Example: Using CSS Selectors

```
<html>
  <head>
    <link rel="stylesheet" href="mystyles.css">
  </head>
  <body>
    <span lang="en">Hello World</span>
    <p>I am learning
      <span id="abbrev">CSS</span>
    </p>
  </body>
</html>
```

```
span {
  border-color: red
}
```

Hello World

I am learning

CSS

# Example: Using CSS Selectors

```
<html>
  <head>
    <link rel="stylesheet" href="mystyles.css">
  </head>
  <body>
    <span lang="en">Hello World</span>
    <p>I am learning
      <span id="abbrev">CSS</span>
    </p>
  </body>
</html>
```

```
[lang] {
  border-color: red
}
```

Hello World

I am learning CSS

# Example: Using CSS Selectors

```
<html>
  <head>
    <link rel="stylesheet" href="mystyles.css">
  </head>
  <body>
    <span lang="en">Hello World</span>
    <p>I am learning
      <span id="abbrev">CSS</span>
    </p>
  </body>
</html>
```

```
[lang] {
  border-color: red
}
```

Hello World

I am learning CSS

# Example: Using CSS Selectors

```
<html>
  <head>
    <link rel="stylesheet" href="mystyles.css">
  </head>
  <body>
    <span lang="en">Hello World</span>
      <p>I am learning
        <span id="abbrev">CSS</span>
      </p>
    </body>
</html>
```

```
[lang=de] {
  border-color: red
}
```

Hello World

I am learning CSS

# Example: Using CSS Selectors

```
<html>
  <head>
    <link rel="stylesheet" href="mystyles.css">
  </head>
  <body>
    <span lang="en">Hello World</span>
    <p>I am learning
      <span id="abbrev">CSS</span>
    </p>
  </body>
</html>
```

```
[lang=de] {
  border-color: red
}
```

Hello World

I am learning CSS

# Example: Using CSS Selectors

```
<html>
  <head>
    <link rel="stylesheet" href="mystyles.css">
  </head>
  <body>
    <span lang="en">Hello World</span>
    <p>I am learning
      <span id="abbrev">CSS</span>
    </p>
  </body>
</html>
```

```
#abbrev {
  border-color: red
}
```

Hello World  
I am learning CSS

```
span {
  border-color: red
}
```

Hello World

I am learning CSS

```
[lang] {
  border-color: red
}
```

Hello World

I am learning CSS

```
[lang=de] {
  border-color: red
}
```

Hello World

I am learning CSS

# CSS Selector Specificity

```
/* In mystyles.css */
```

```
div {  
    background: red;  
}
```

```
#top {  
    background: green;  
}
```

```
.warn {  
    background: yellow;  
}
```

```
<!-- In HTML -->
```

```
<div id="top" class="warn">  
    Sample  
</div>
```



Which selector wins?

# CSS Selector Specificity

```
/* In mystyles.css */  
div {  
    background: red;  
}  
  
#top {  
    background: green;  
}  
  
.warn {  
    background: yellow;  
}
```

```
<!-- In HTML -->  
<div id="top" class="warn">  
    Sample  
</div>
```



## Which selector wins?

Rule1:

IDs - classes/attributes - elements



# CSS Selector Specificity

```
p {  
  color: black;  
}  
/* 0-0-1 */  
  
.notice {  
  color: blue;  
}  
/* 0-1-0 */  
  
.notice.important {  
  color: orange;  
}  
/* 0-2-0 */  
  
p.important {  
  color: orange;  
}  
/* 0-1-1 */
```

```
<div id="app">  
  <p class="notice important">Warning</p>  
</div>
```



## Which selector wins?

Rule1:

IDs - classes/attributes - elements

Rule2: Compare number of selectors if they are at the same specificity

# CSS Selector Specificity

```
p {  
  color: black;  
}  
/* 0-0-1 */  
  
.notice {  
  color: blue;  
}  
/* 0-1-0 */  
  
.notice.important {  
  color: orange;  
}  
/* 0-2-0 */  
  
p.important {  
  color: orange;  
}  
/* 0-1-1 */
```

```
<div id="app">  
  <p class="notice important">Warning</p>  
</div>
```



## Which selector wins?

Rule1:

IDs - classes/attributes - elements

Rule2: Compare number of selectors if they are at the same specificity

Rule3: If three-number notations are exactly the same, choose the one that is defined later

# CSS Selector Specificity Exercise

```
<div id="header">  
  <ul class="menu">  
    <li class="item">Home</li>  
    <li class="item">About</li>  
    <li class="item special">Services</li>  
    <li class="item">Contact</li>  
  </ul>  
</div>
```

HTML



What is the color of the "Services"?

Hints:

Let's assume all four selectors can all select the "Services" content

CSS

```
/* Selector 1 */  
ul.menu li {  
  color: blue;  
}  
  
/* Selector 2 */  
li.item {  
  color: green;  
}  
  
/* Selector 3 */  
#header .item.special {  
  color: red;  
}  
  
/* Selector 4 */  
#header ul li.special {  
  color: orange;  
}
```

Practice